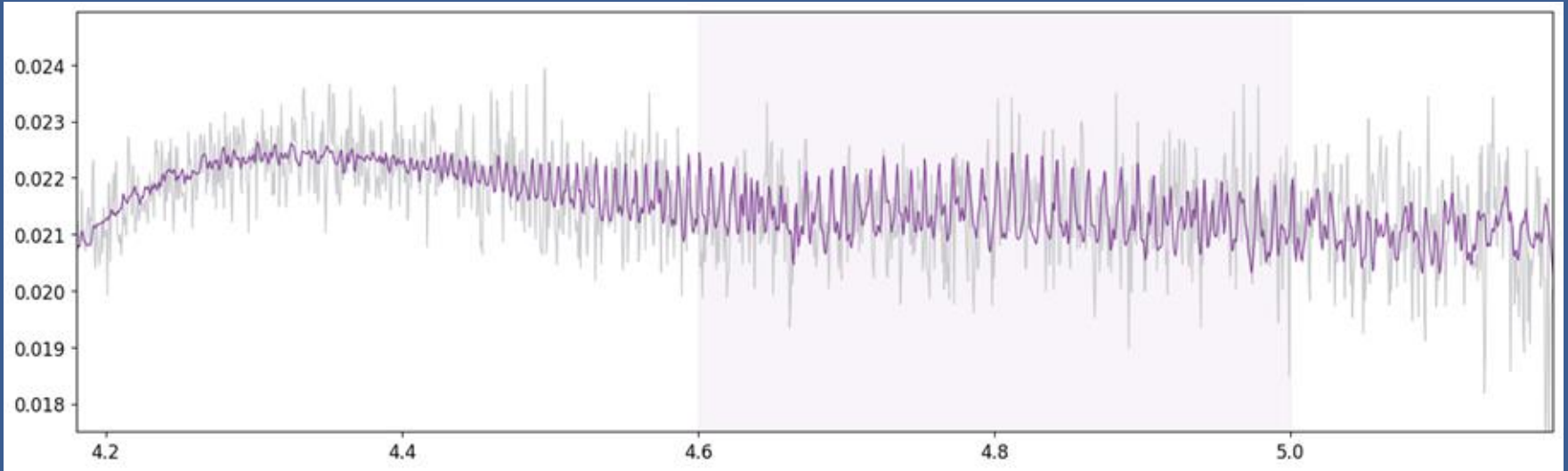


Retrieving Carbon Monoxide Isotopes in JWST Transmission Spectra: A Case Study with WASP-39b



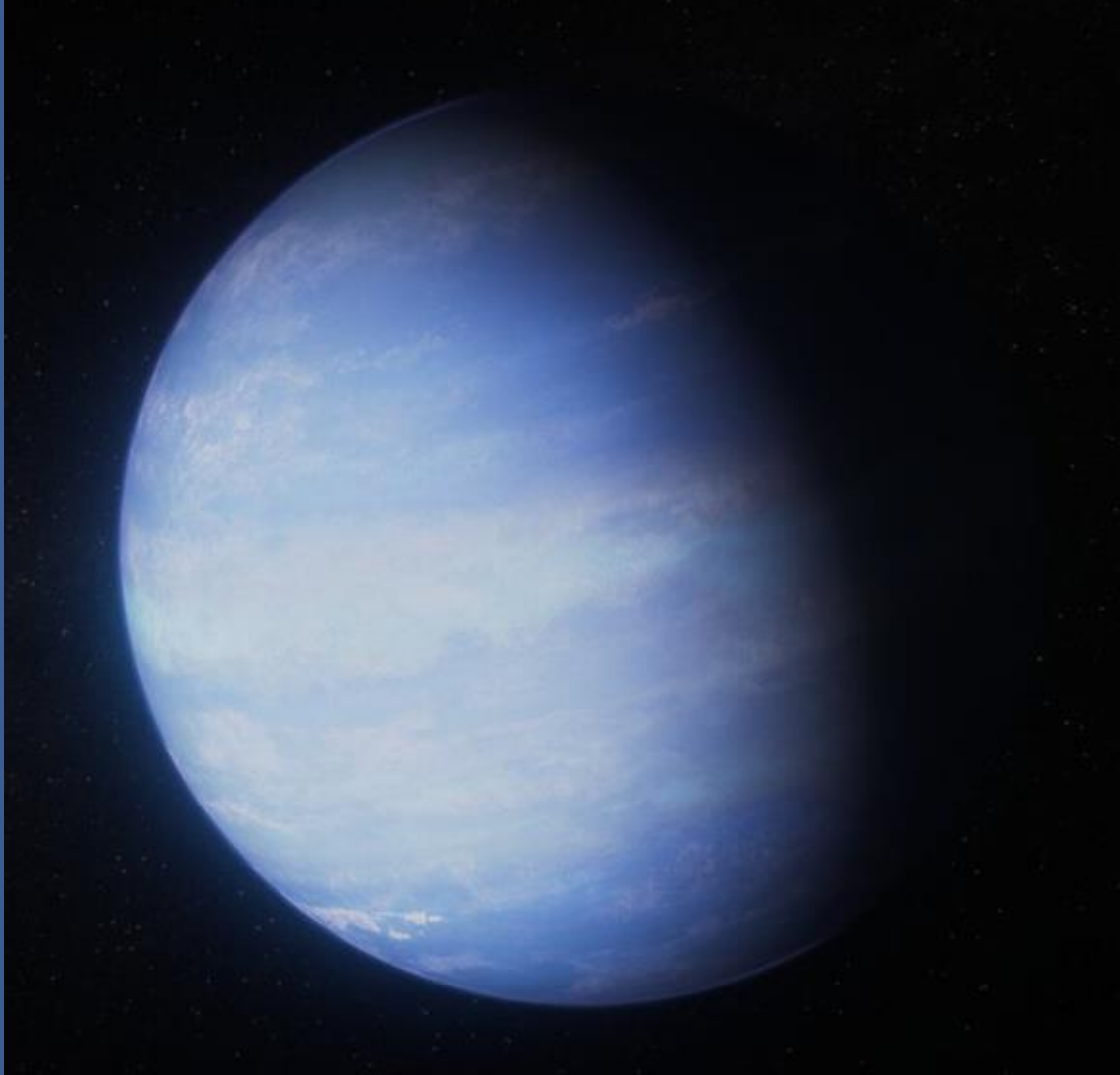
Outline

Introduction — What we can learn from isotopes in exoplanet atmospheres

Background — A hint of carbon monoxide isotopes in WASP-39b

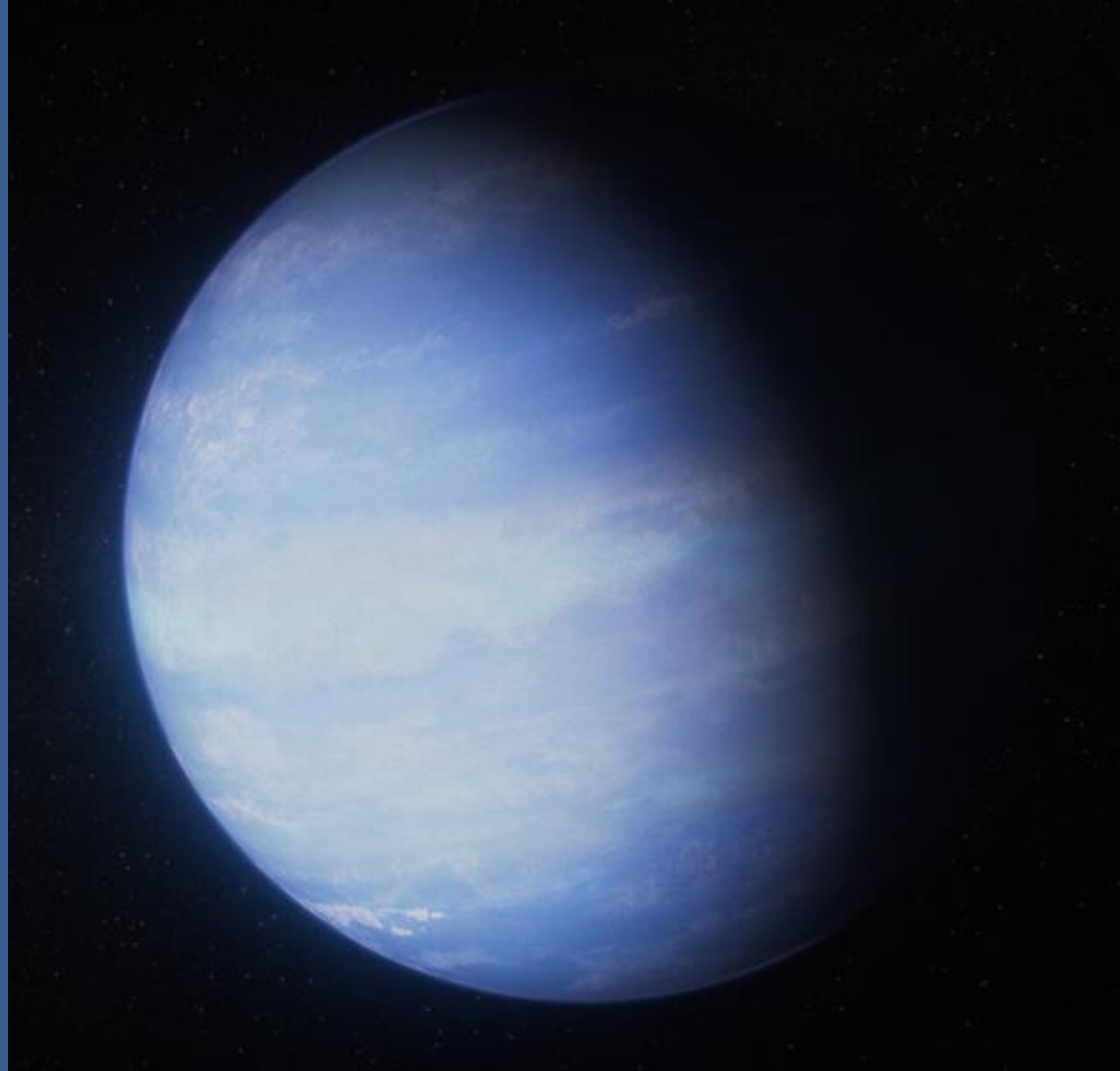
Methods — Running retrievals on WASP-39b

Results — Carbon monoxide isotope ratios and biases



Introduction

What can we learn from isotopes in exoplanet atmospheres?



Why do we study exoplanet atmospheres?

Exoplanet atmospheres are a window into a planet's atmospheric and overall composition.

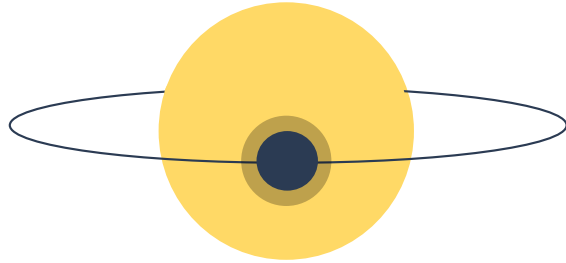
Factors that shape composition:

- **planet formation** (formation location, composition of protoplanetary disk)
- **planet evolution** (interior and surface processes)
- **stellar irradiation** (heating, atmospheric escape)
- **atmospheric chemistry** (chemical processes, winds, circulation)

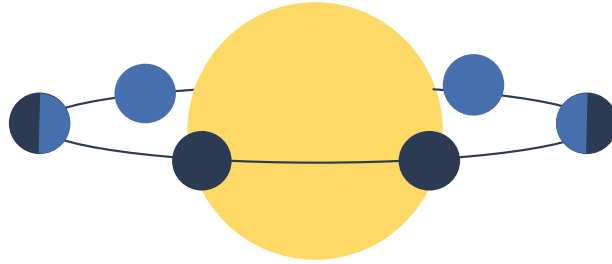


How do we study exoplanet atmospheres?

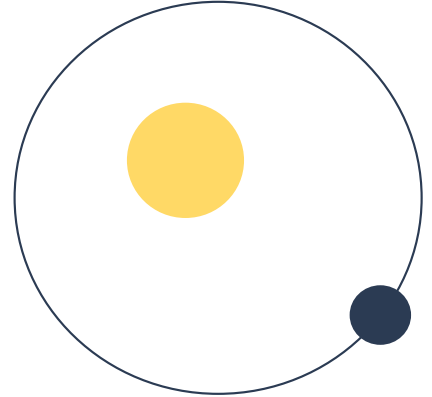
transmission spectroscopy



phase curves



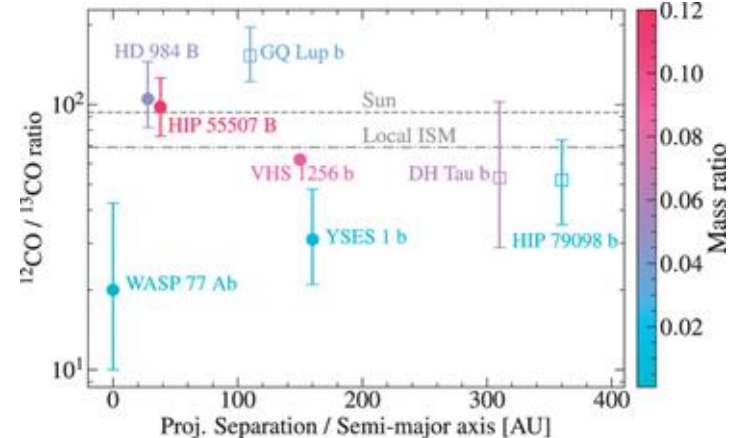
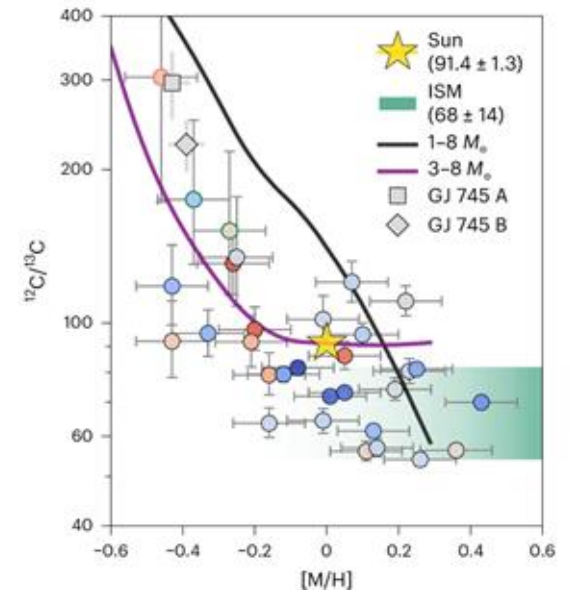
high-contrast spectroscopy



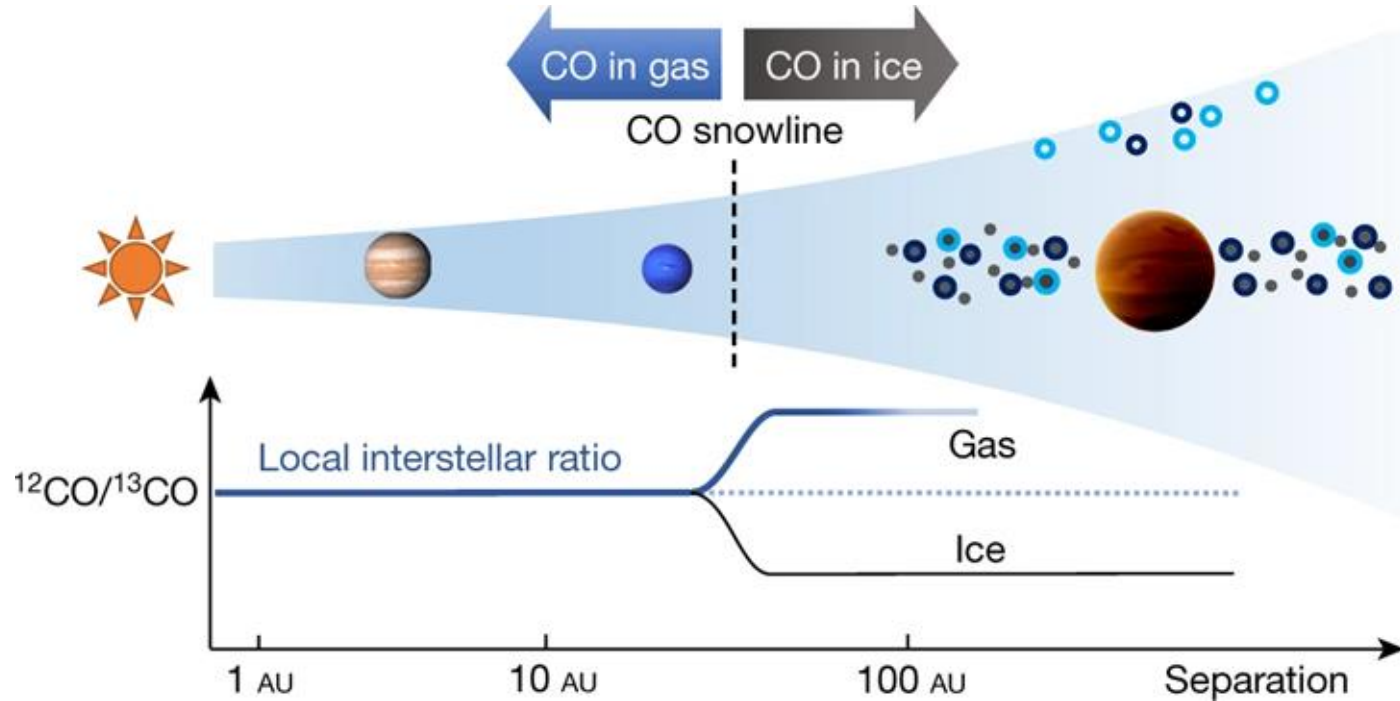
$^{12}\text{C}/^{13}\text{C}$ Isotope Ratios

The ratios between different isotopes probe chemical evolution in the universe

We are just beginning to measure $^{12}\text{C}/^{13}\text{C}$ ratios for a large population of brown dwarfs and planets — gleaming insights into composition



What can $^{12}\text{CO}/^{13}\text{CO}$ tell us about planet formation?



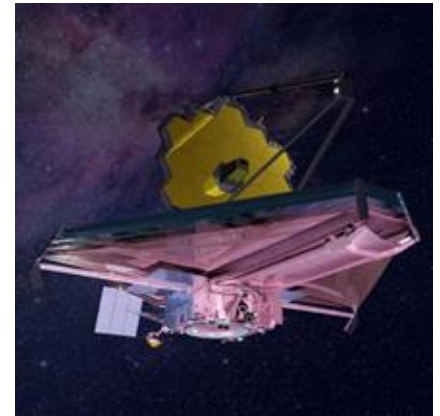
We can try measuring carbon monoxide isotopes with JWST!

Previous work measured CO isotopes with:

→ instruments with $R \sim 30,000$ -100,000
(high-resolution)

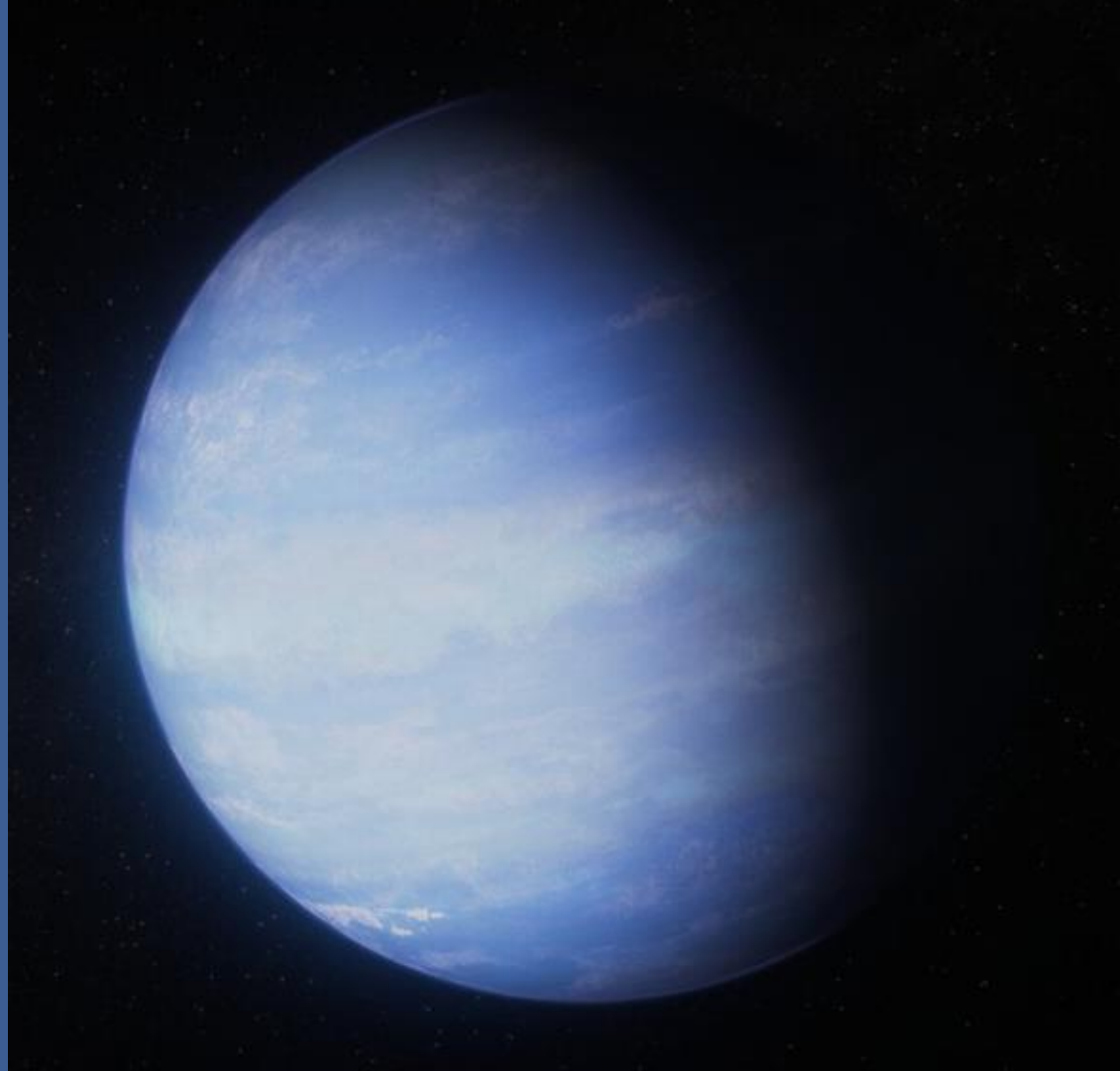
→ high-contrast imaging or phase
curves, which have **higher signal-to-
noise** (S/N) than transmission

We aim to measure CO isotopes at medium
resolution and low S/N, with JWST NIRSpec
G395H ($R \sim 3,000$ -4,000)



Background

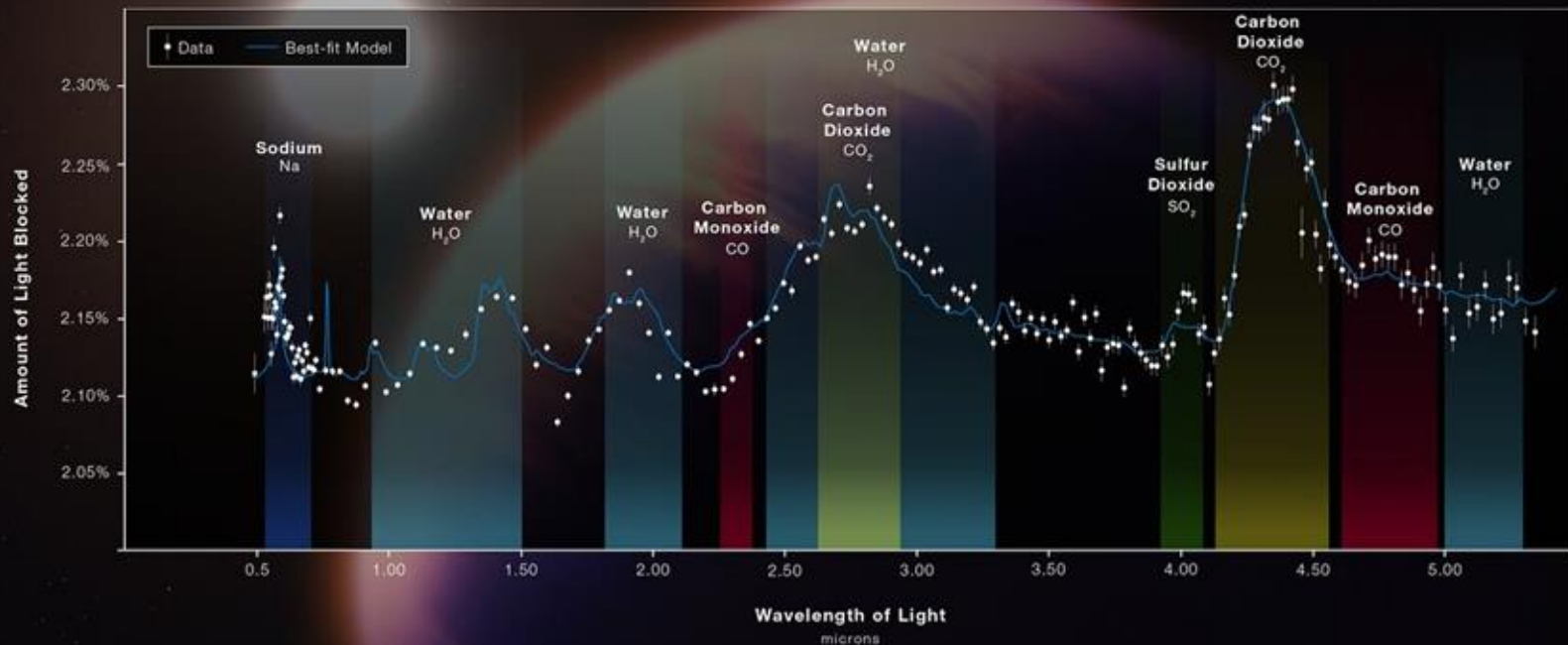
A hint of carbon monoxide
isotopes in WASP-39b



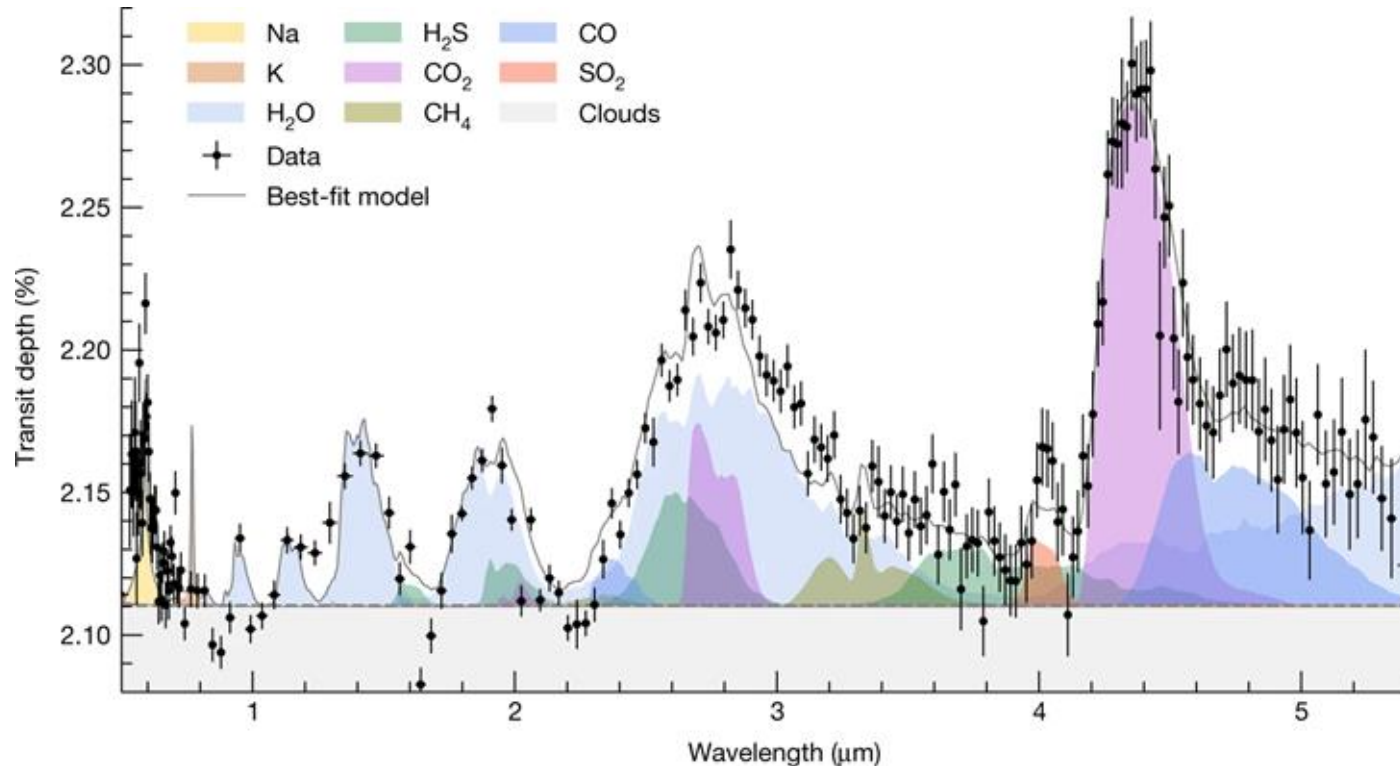
HOT GAS GIANT EXOPLANET WASP-39 b

ATMOSPHERE COMPOSITION

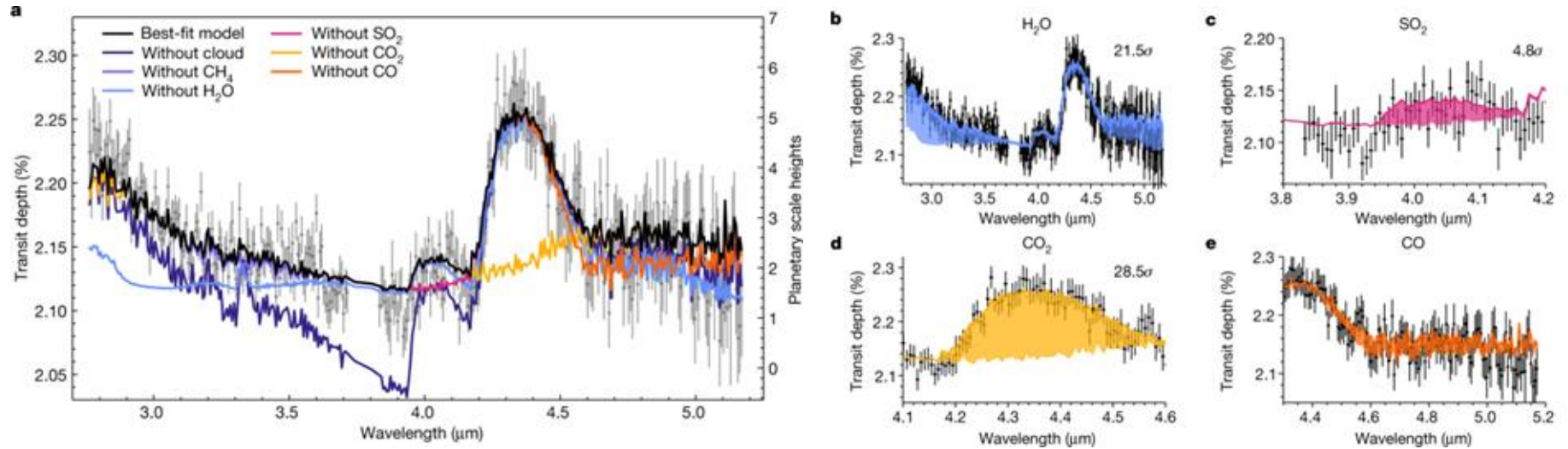
NIRSpec PRISM



Detected several molecular species

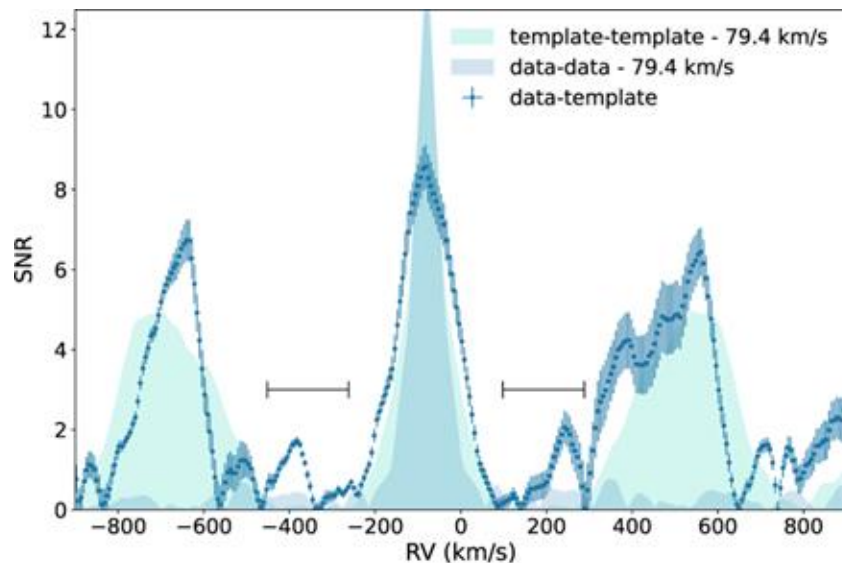


CO was previously detected in 4.7 micron band of NIRSpec G395H (but not significant)



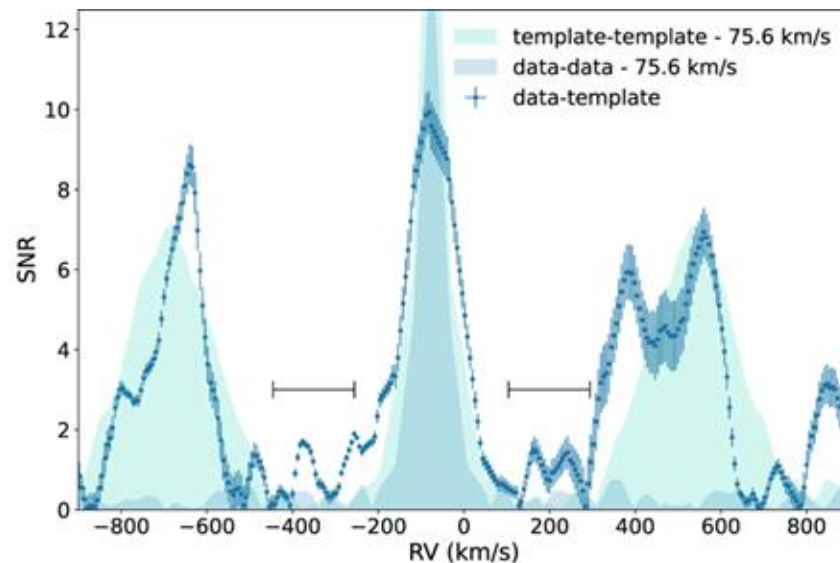
Data was binned down by 10-pixel bins

Esparza-Borges et al. 2023 confirmed the presence of CO with cross-correlation at native pixel resolution → more significant when isotopes were included



(a) CO main isotopologue

CO main isotopologue → 6.6 sigma



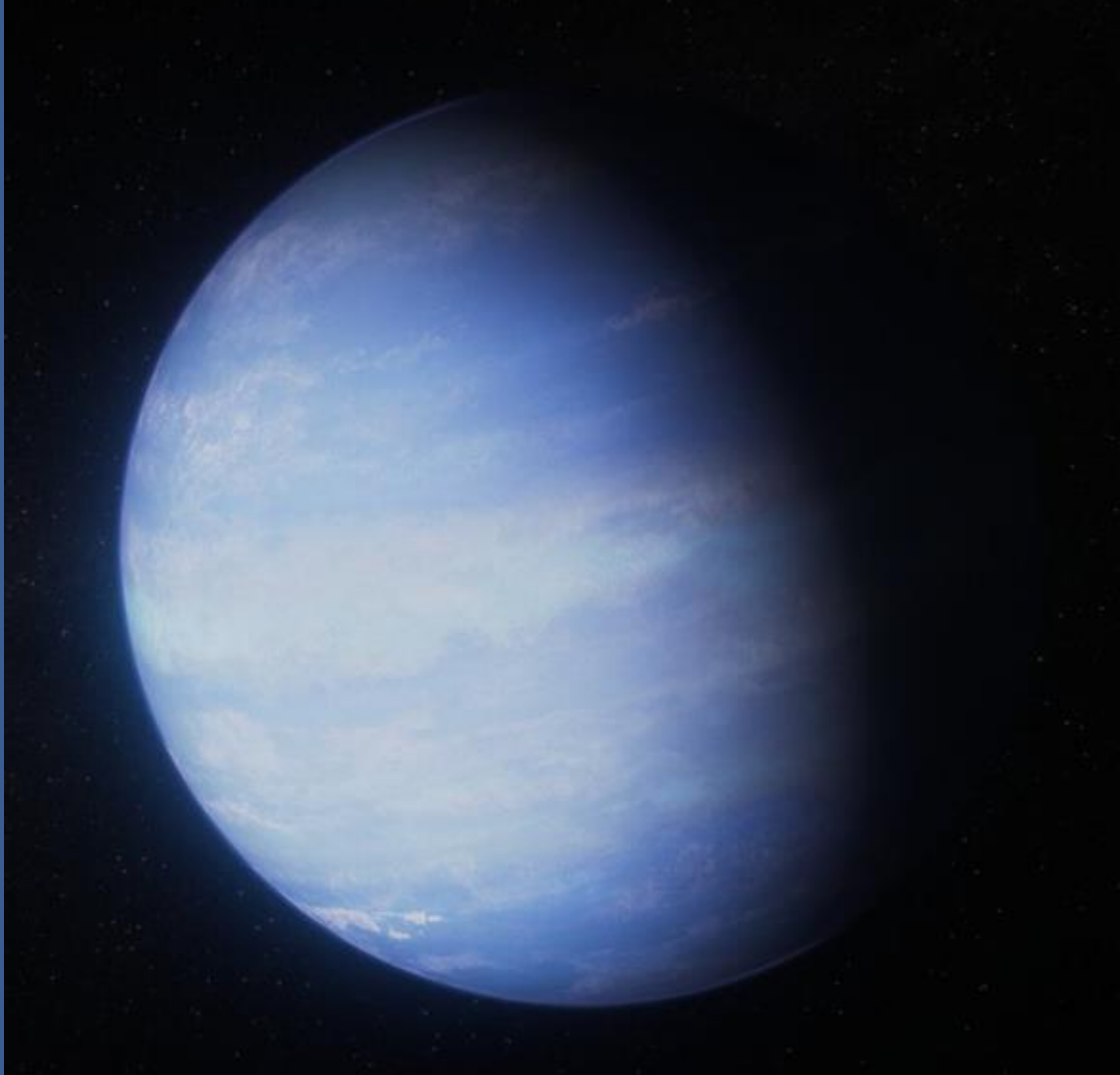
(c) CO all isotopologues

CO all isotopologues → 7.5 sigma

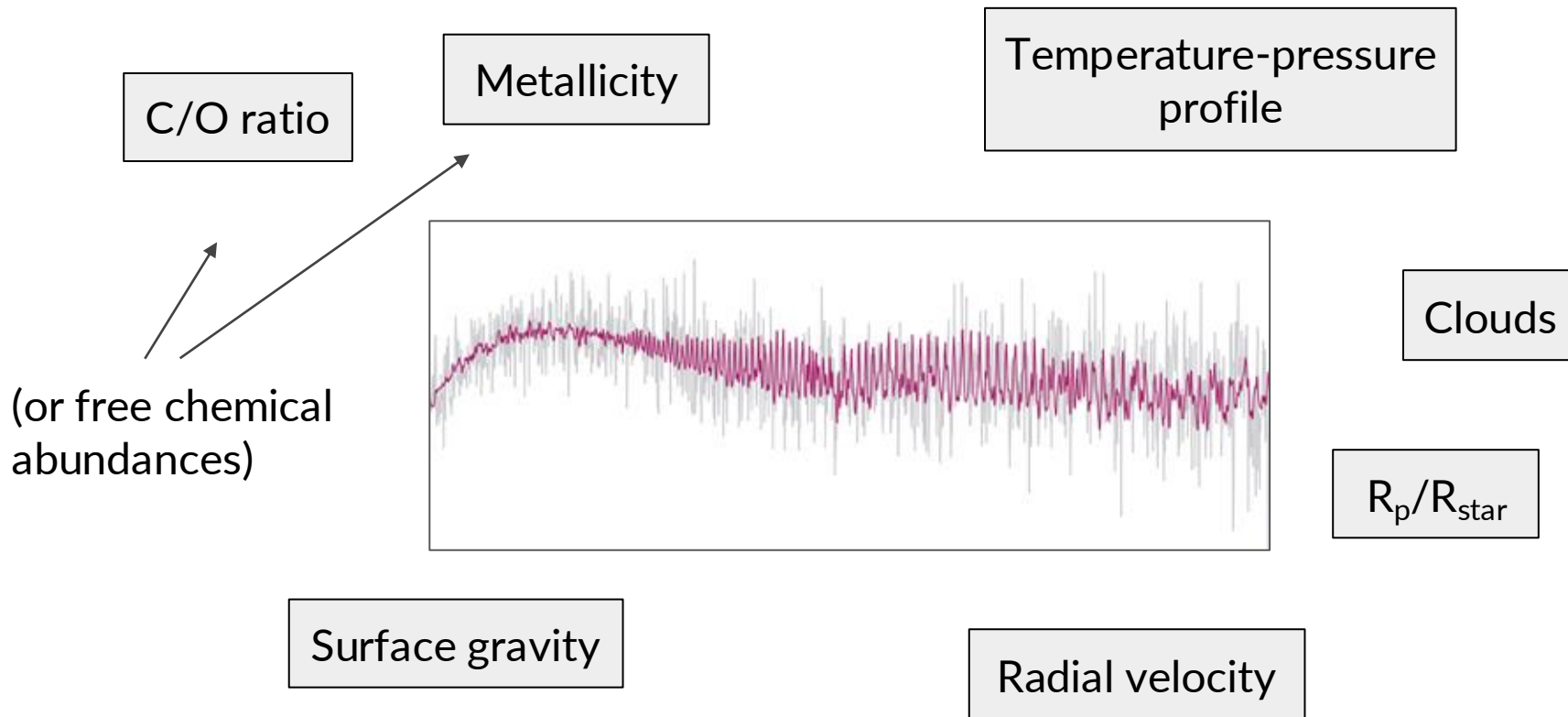
**We aim to fit for the $^{12}\text{CO}/^{13}\text{CO}$ isotope ratio
directly by implementing atmospheric retrievals**

Methods

Running retrievals on
WASP-39b

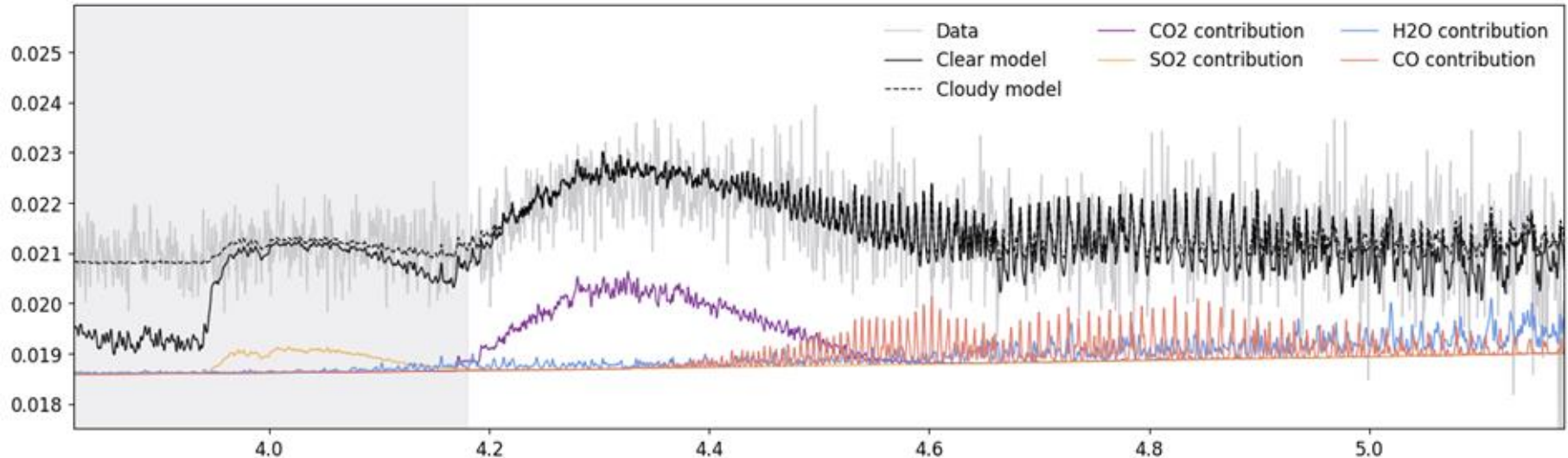


The ingredients of an exoplanet atmosphere



My set-up

- Using data from Esparza-Borges et al. 2023 (JWST NIRSpec G395H)
- Equilibrium chemistry (species: CO₂, CO, H₂O)
- Fit over 4.18–5.3 micron (to exclude SO₂ and clouds, which primarily affect the spectrum from 3.8–4.18 micron)



Retrieval set-up

petitRADTRANS
radiative transfer code
is used to generate the
model spectra

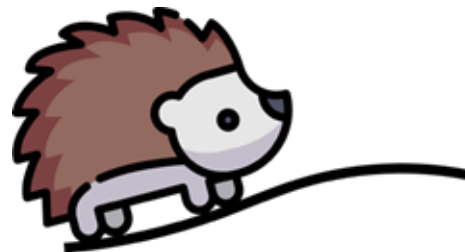


Table 1. WASP-39b Retrieval Priors

Parameter	Prior
T_{iso} (K)	$\mathcal{U}[500, 2000]$
R_p (R_{jup})	$\mathcal{U}[1.1, 1.4]$
C/O	$\mathcal{U}[0.1, 0.9]$
$\log(^{12}\text{CO}/^{13}\text{CO})$	$\mathcal{U}[-1, 4]$
[Fe/H] (dex)	[3, 5, 10]
RV (km/s) ¹	-87.33
g (cm/s ⁻¹) ¹	407.38
$R_\star$ ($R_\odot$) ²	0.895

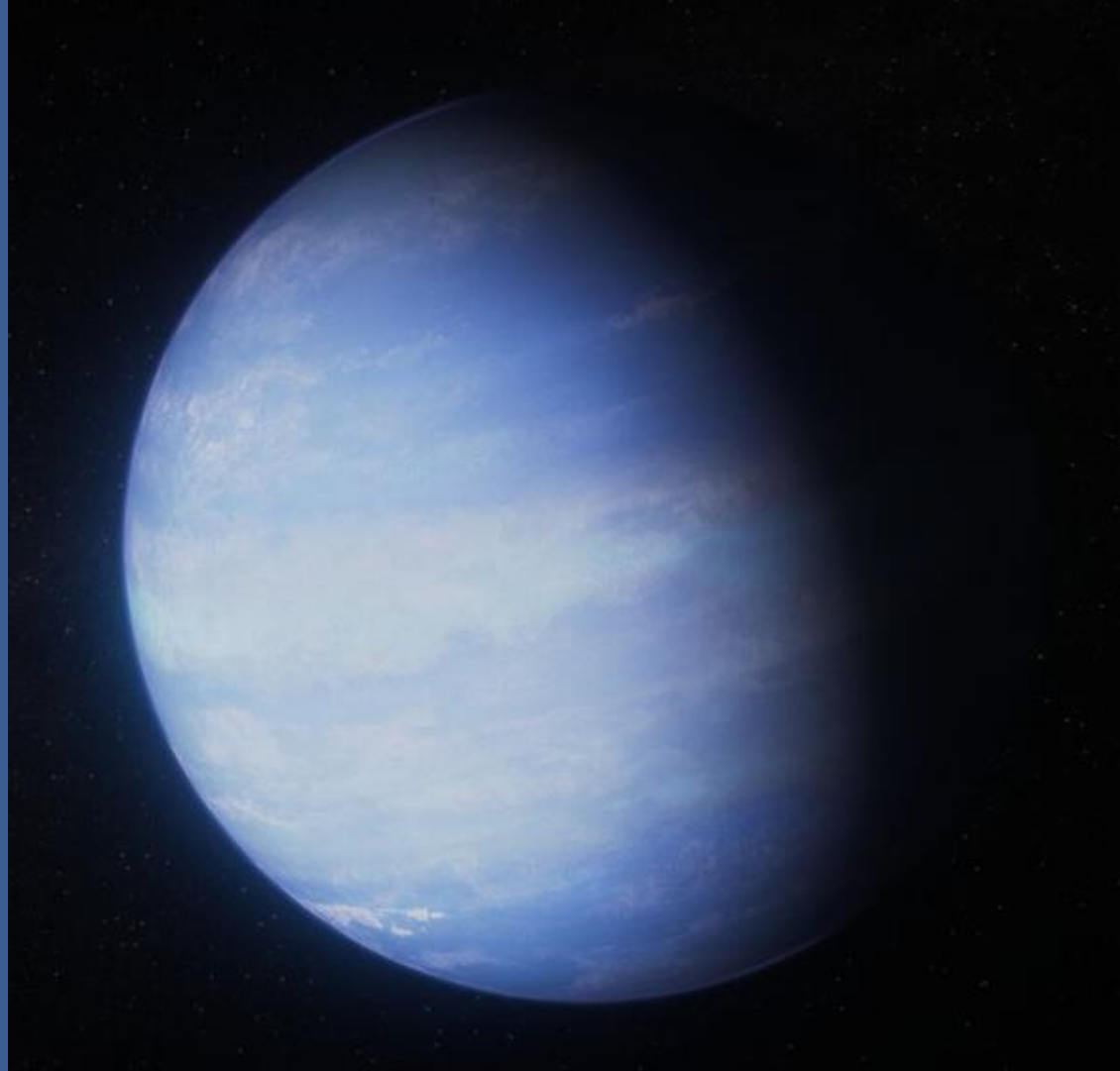
NOTE— $\mathcal{U}[c, d]$ indicates a uniform distribution bounded inclusively between c and d . References: ¹F. Faedi et al. (2011), ²L. Mancini et al. (2018).

UltraNest is used to
implement nested
sampling and find the
most likely parameters

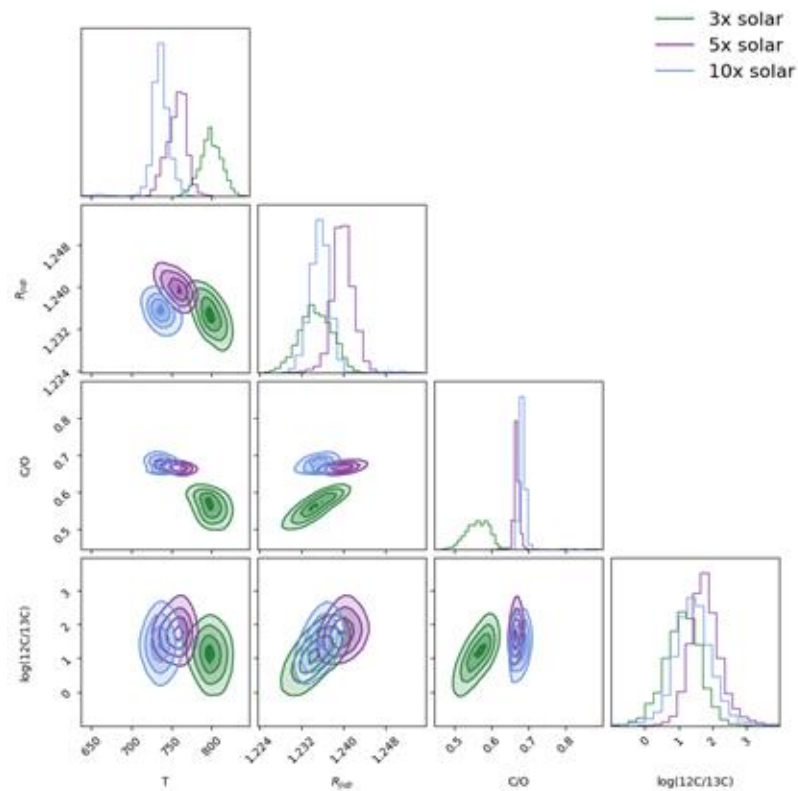
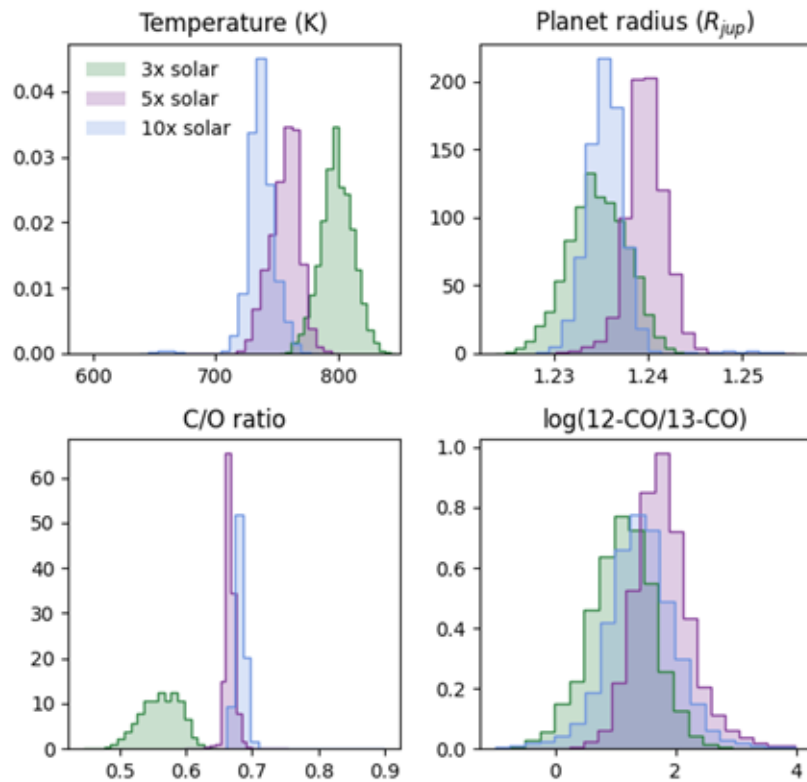


Results

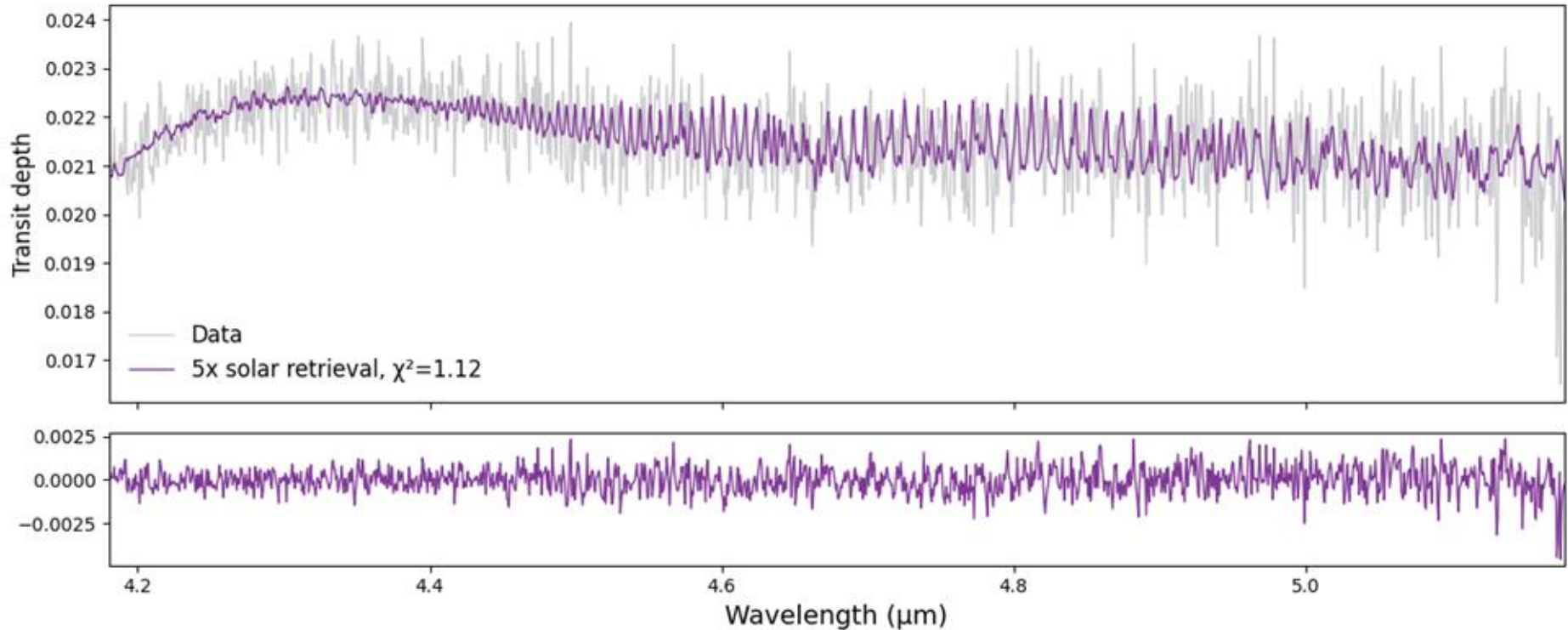
Carbon monoxide isotope
ratios and biases



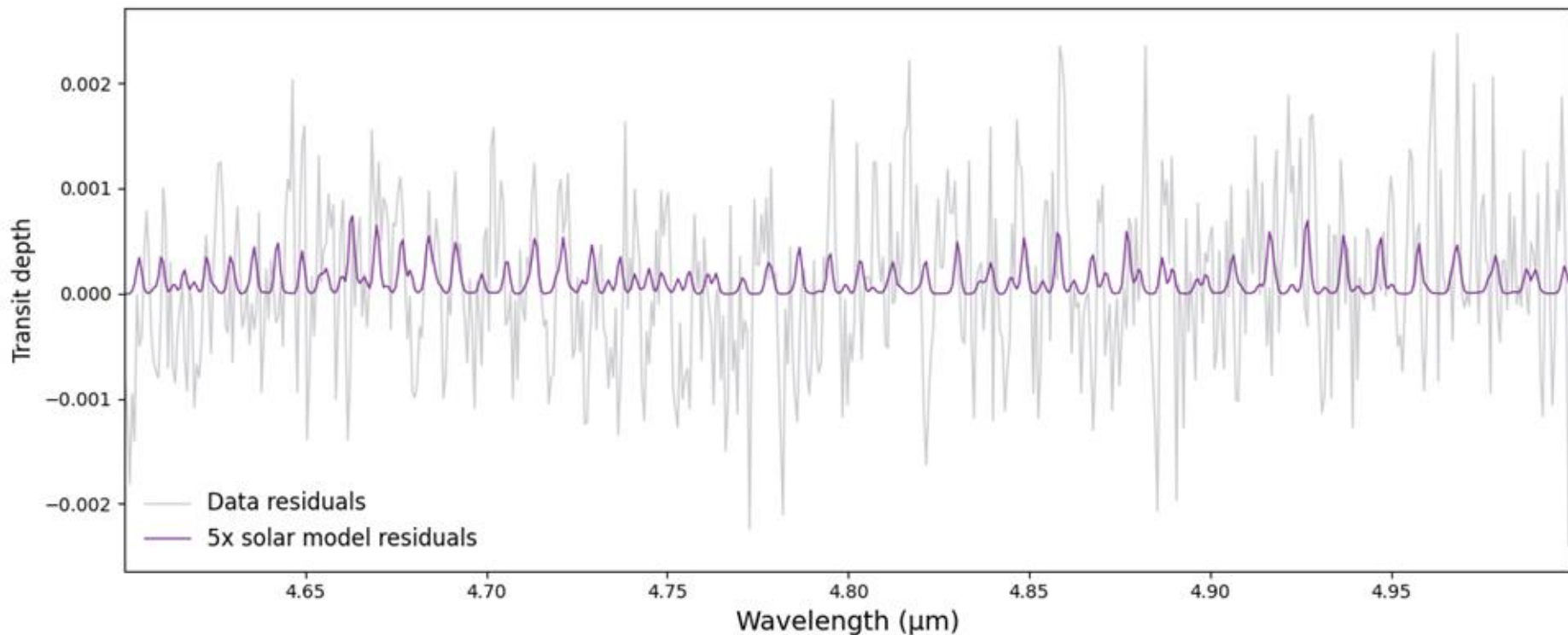
Retrieval posteriors



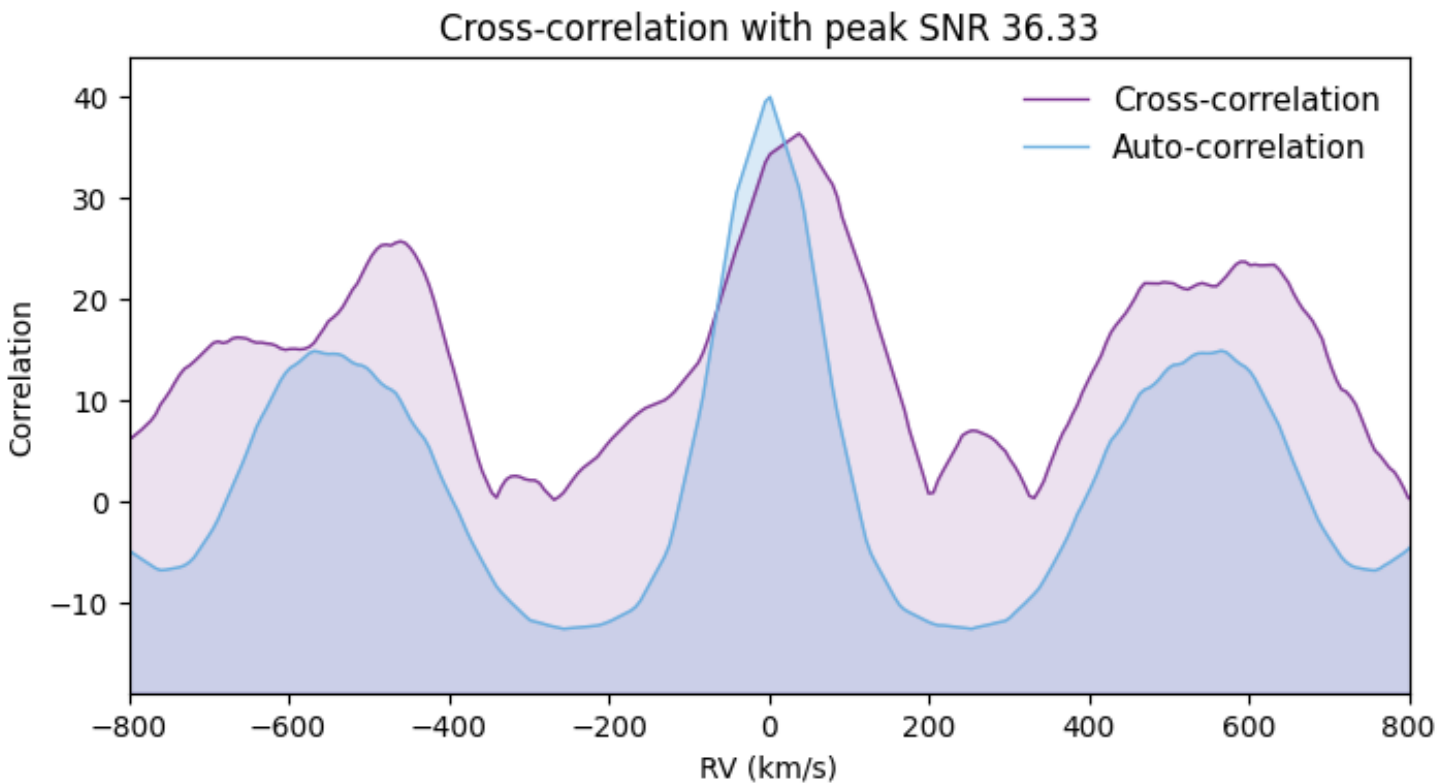
Best fit spectrum (5x solar)



Comparing ^{13}CO residuals (data vs. model)



Cross correlation



Isotope ratios

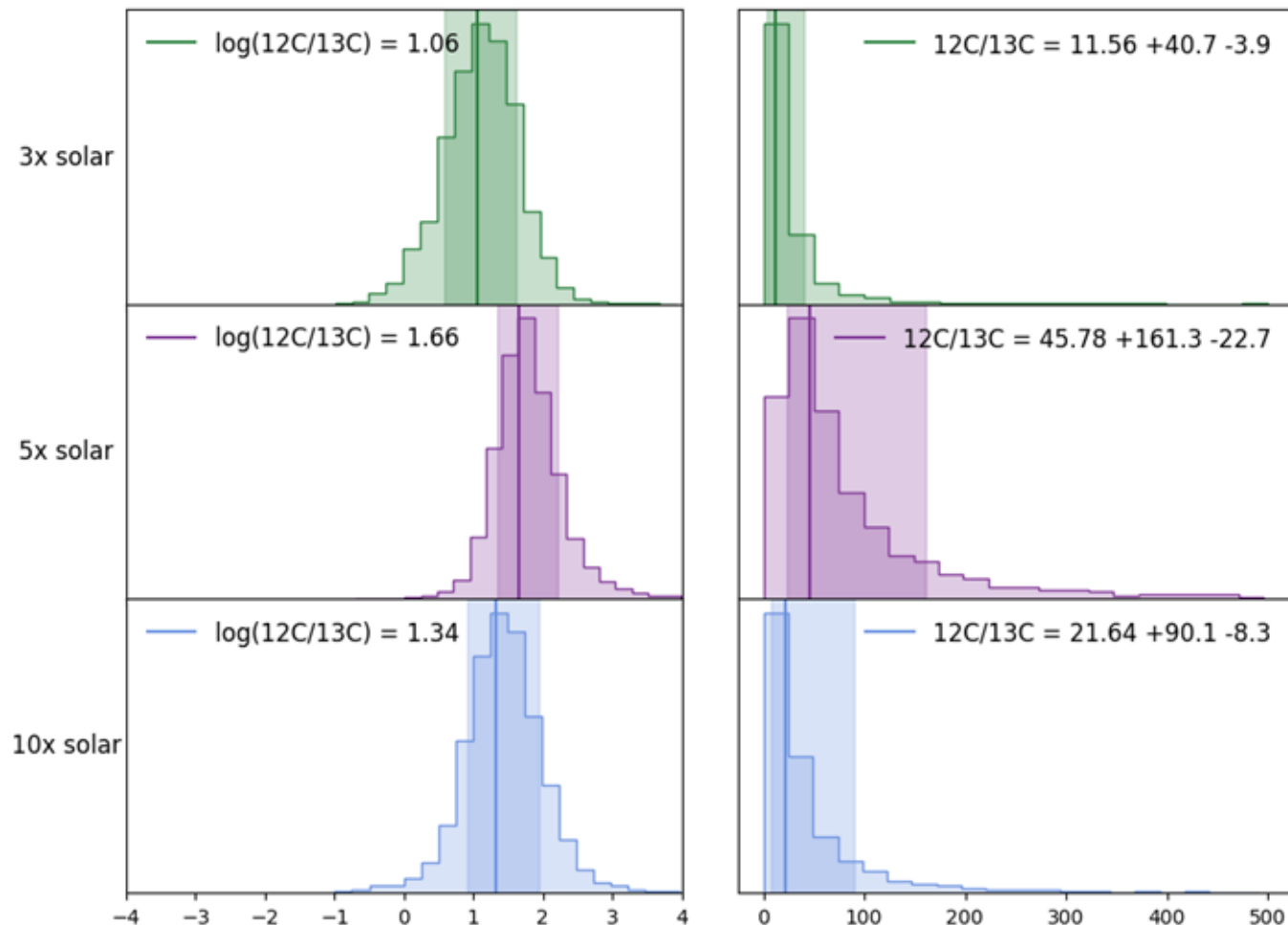
Very low isotope ratios

Finding a lot of ^{13}CO

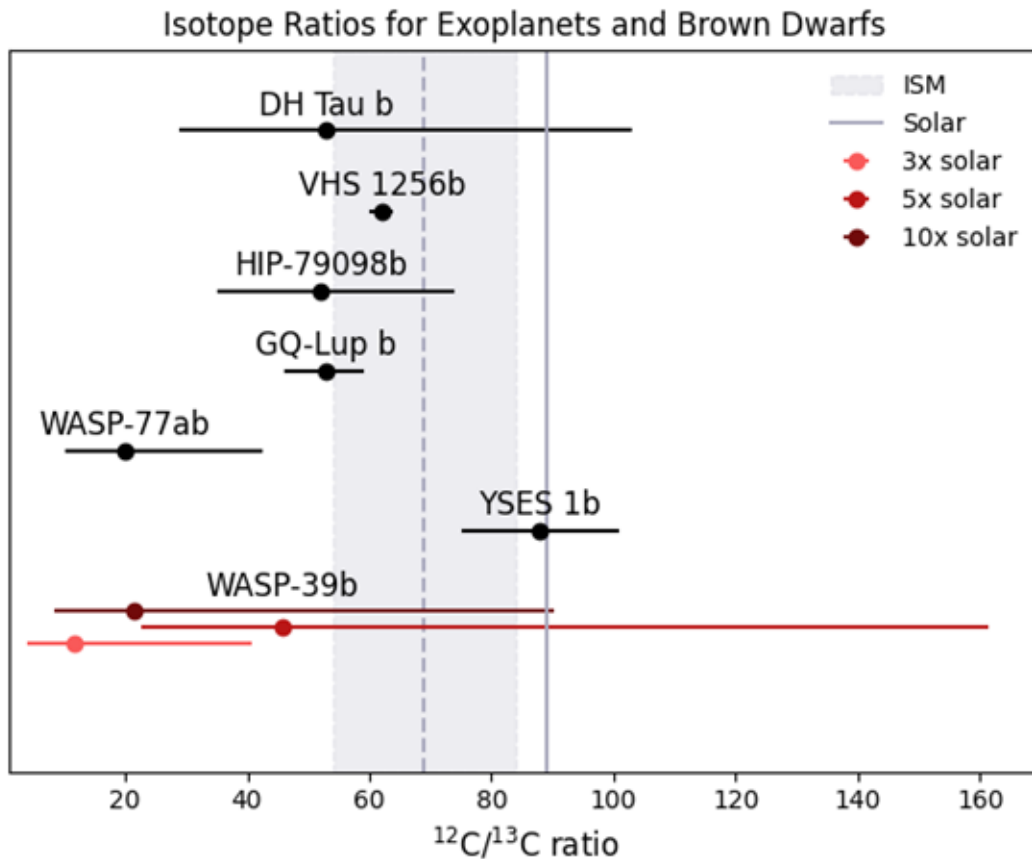
For reference:

ISM $\rightarrow 69 \pm 15$

Solar $\rightarrow 89$



Comparing to other isotope ratios



Line et al. 2021

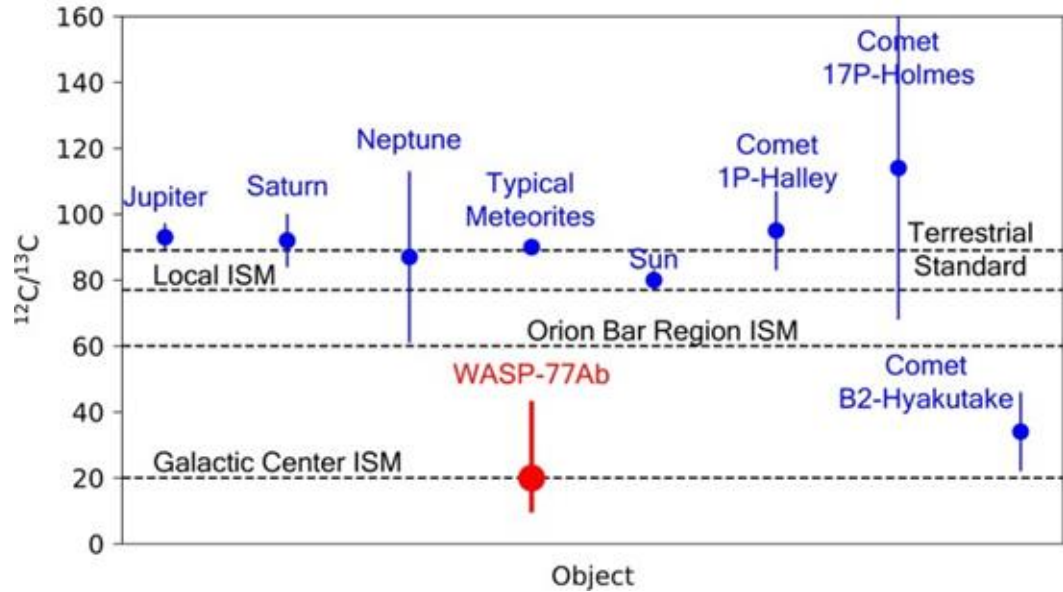
Fits for a $^{12}\text{CO}/^{13}\text{CO}$ isotope ratio from a dayside spectrum of WASP-77Ab

$R \sim 45,000$ (SNR > 200)

Wavelength band: 1.43–2.42 micron

Also find a very low isotope ratio:

$^{12}\text{CO}/^{13}\text{CO} = 10.2\text{--}42.6$

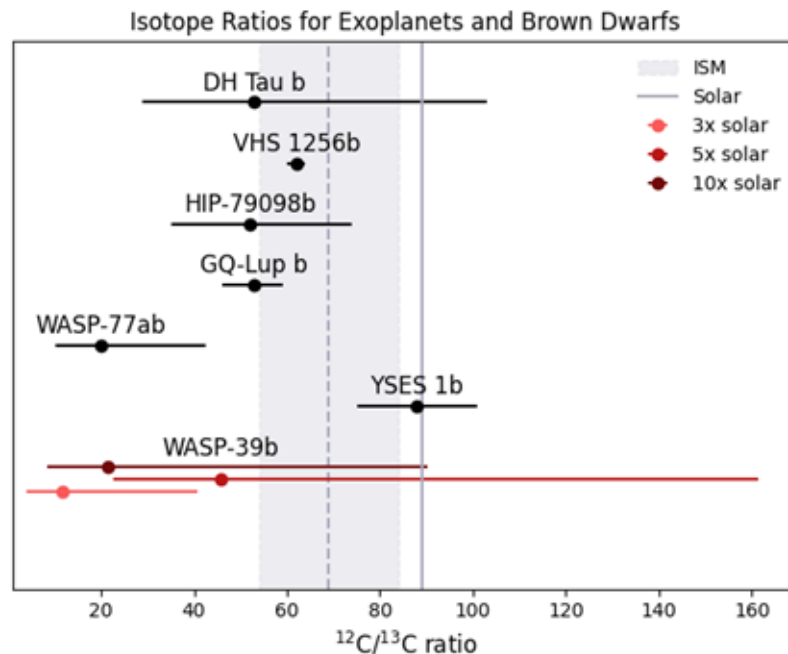


Conclusions

We can fit for the isotope ratio of WASP-39b, but the uncertainties are very large at low S/N

We detect ^{13}CO when we cross-correlate the residuals

More observations (increasing S/N) can help narrow the range of the isotope ratio

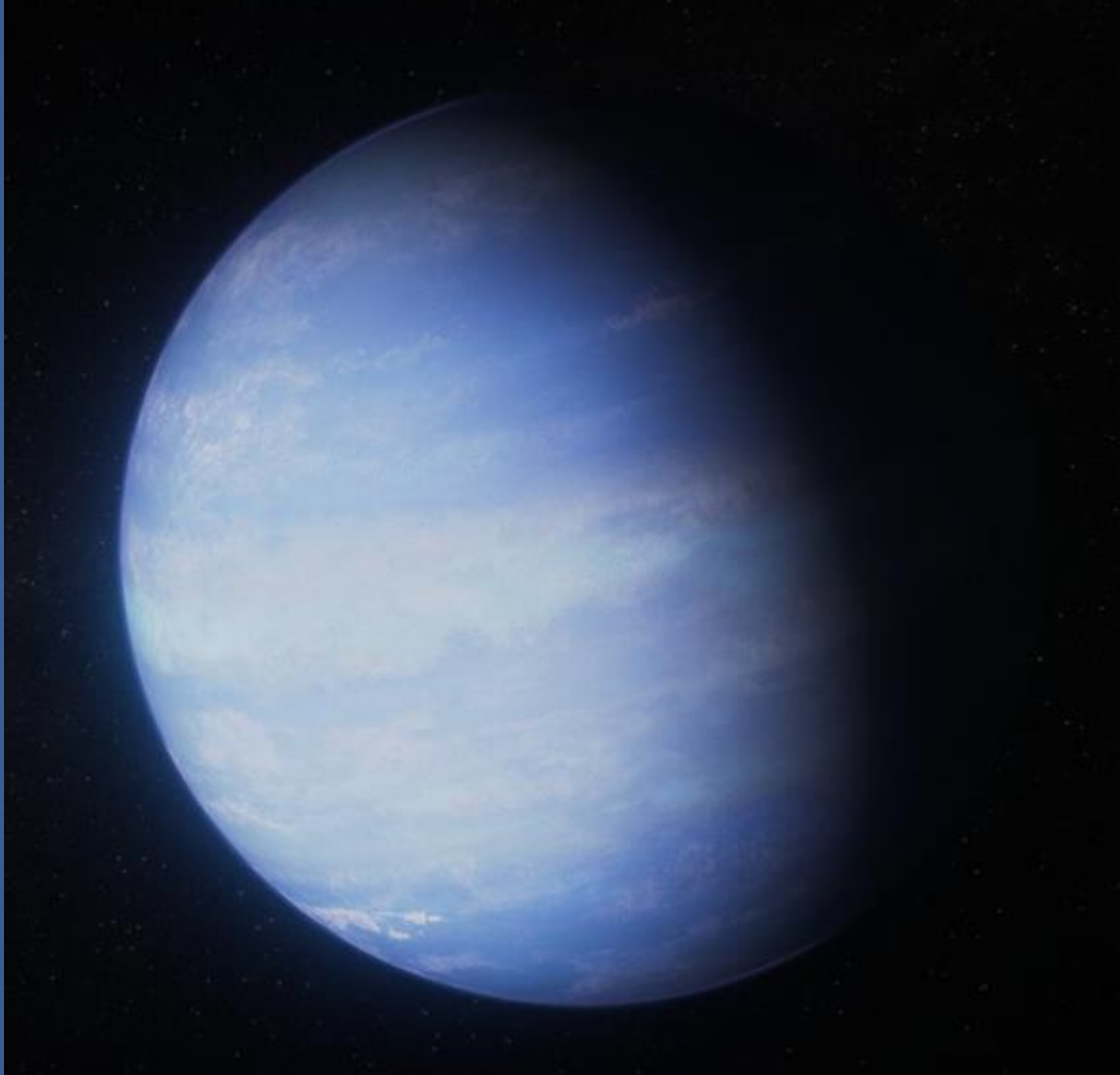


Questions?

Thank you to Ignas Snellen for his invaluable guidance and advice, to Ignas's group (Sam, Dario, Natalie, Lars, Thomas, Chenyang) for their insights every week, and to Leiden Observatory for graciously hosting me for the year!



Universiteit
Leiden



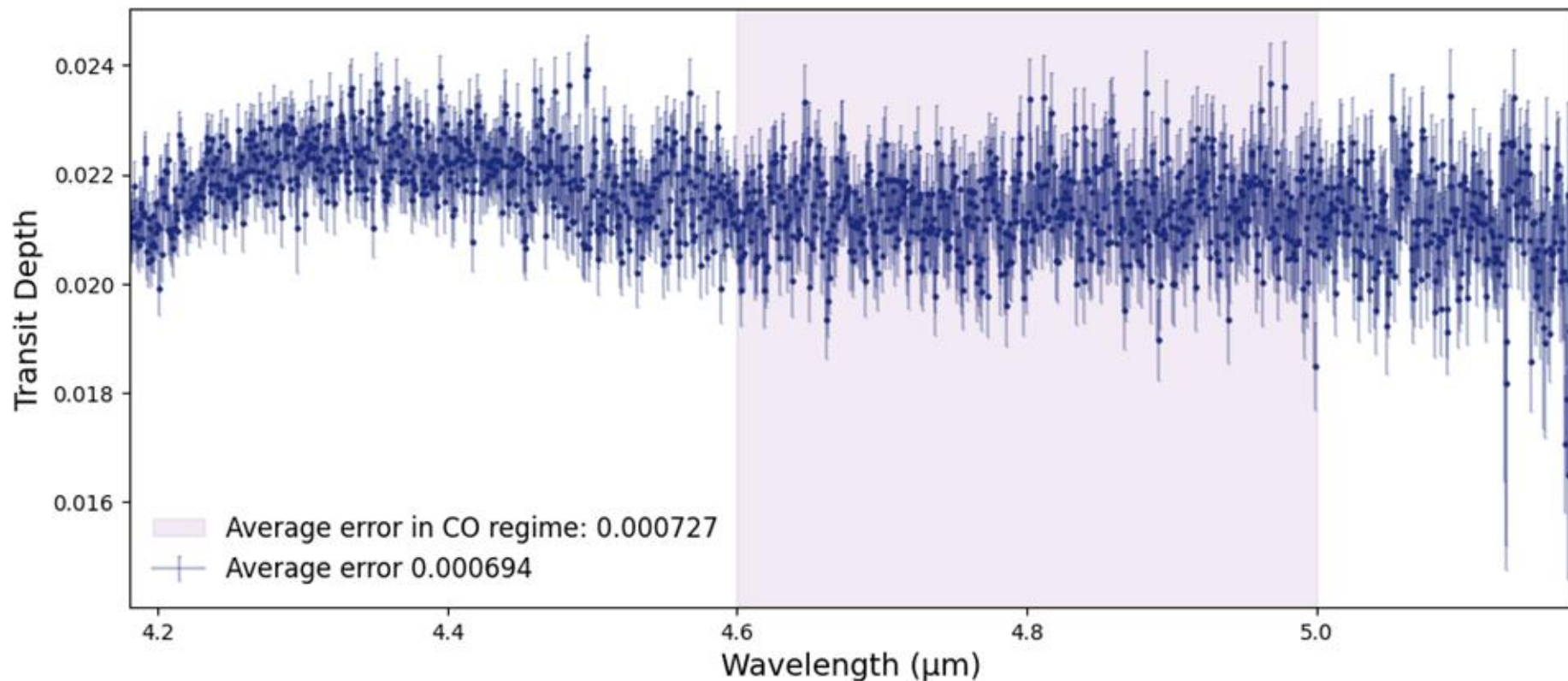
Extra slides

How does noise impact the outcome of retrievals?

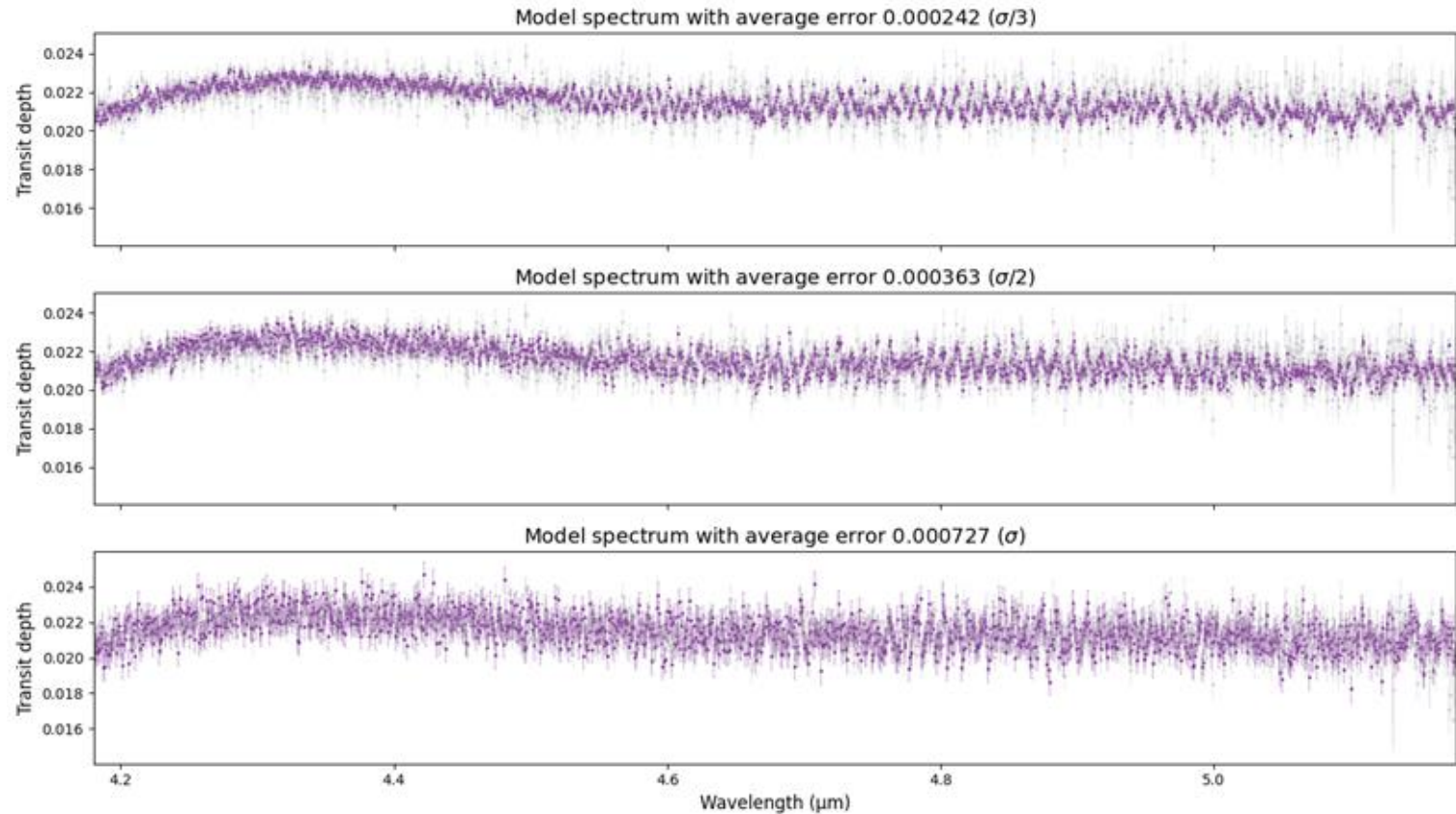
Testing different parameterizations of the isotope:

1. Fitting for $\log(^{12}\text{CO}/^{13}\text{CO})$
2. Fitting for $\log(^{13}\text{CO})$ abundance directly

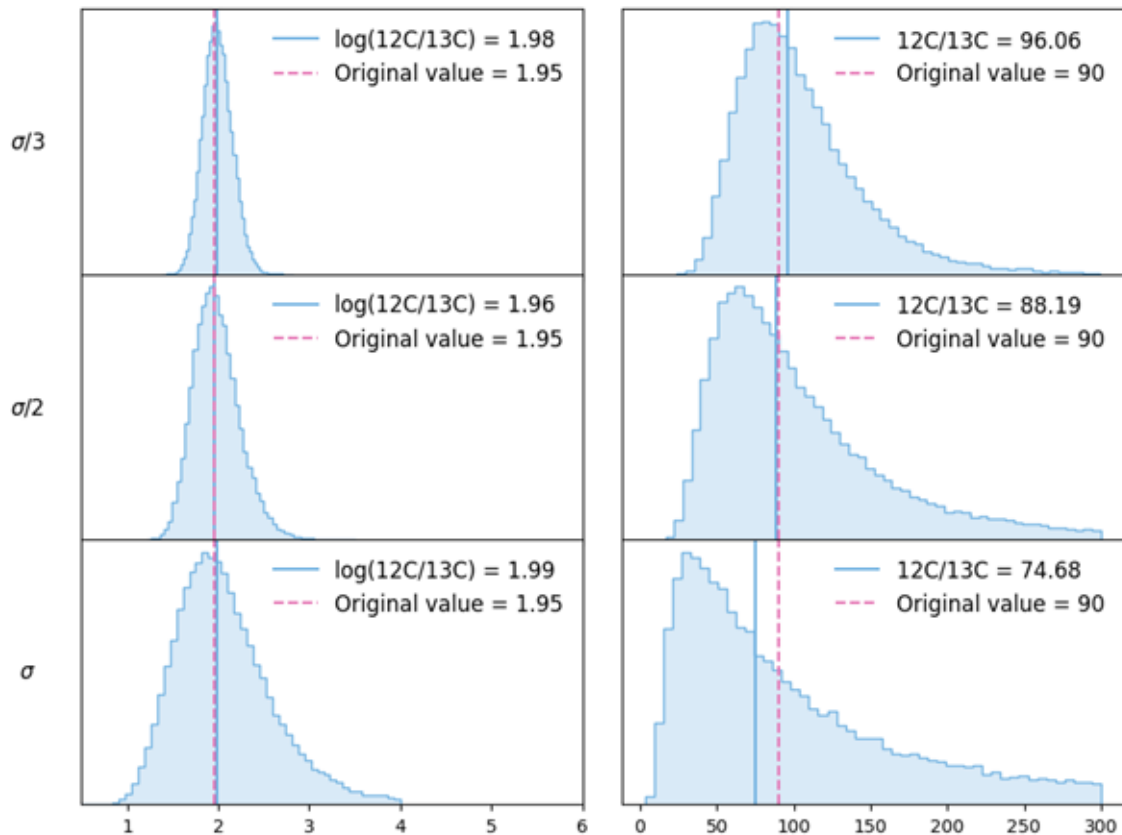
Adding noise (average error = 0.000727)



Simulated noisy models



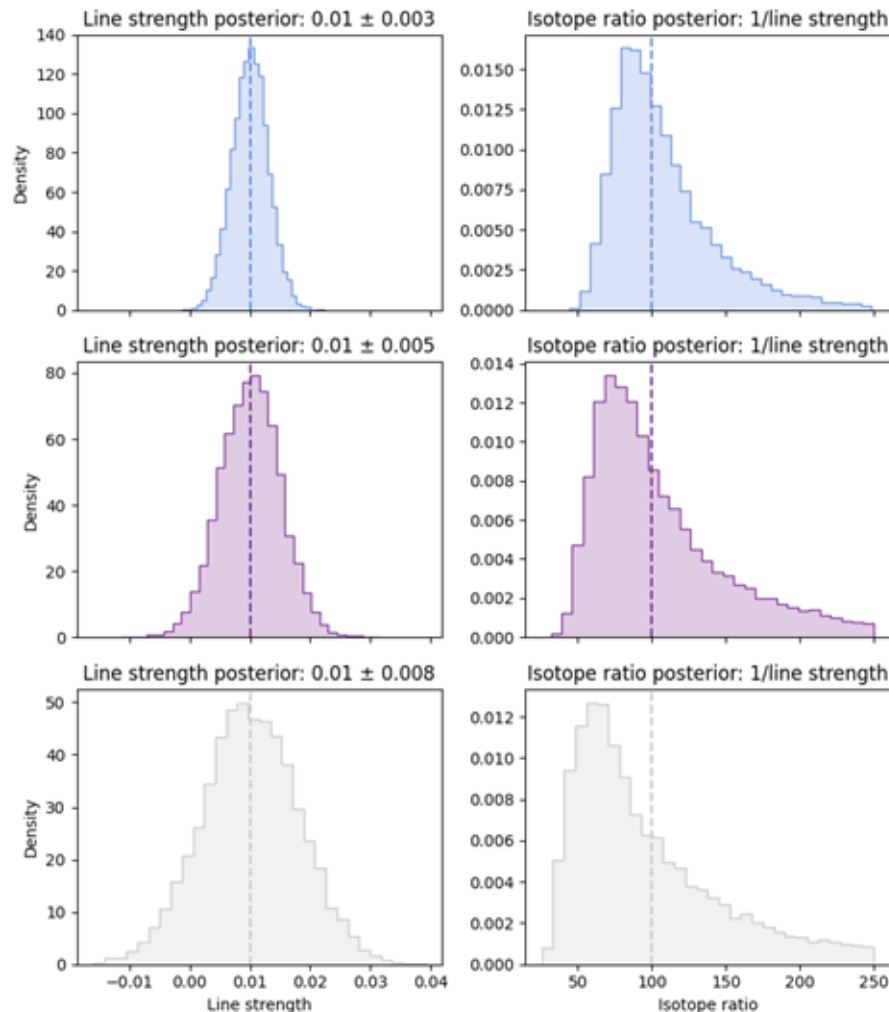
Fitting for isotope ratio $\rightarrow {}^{12}\text{C}/{}^{13}\text{C}$



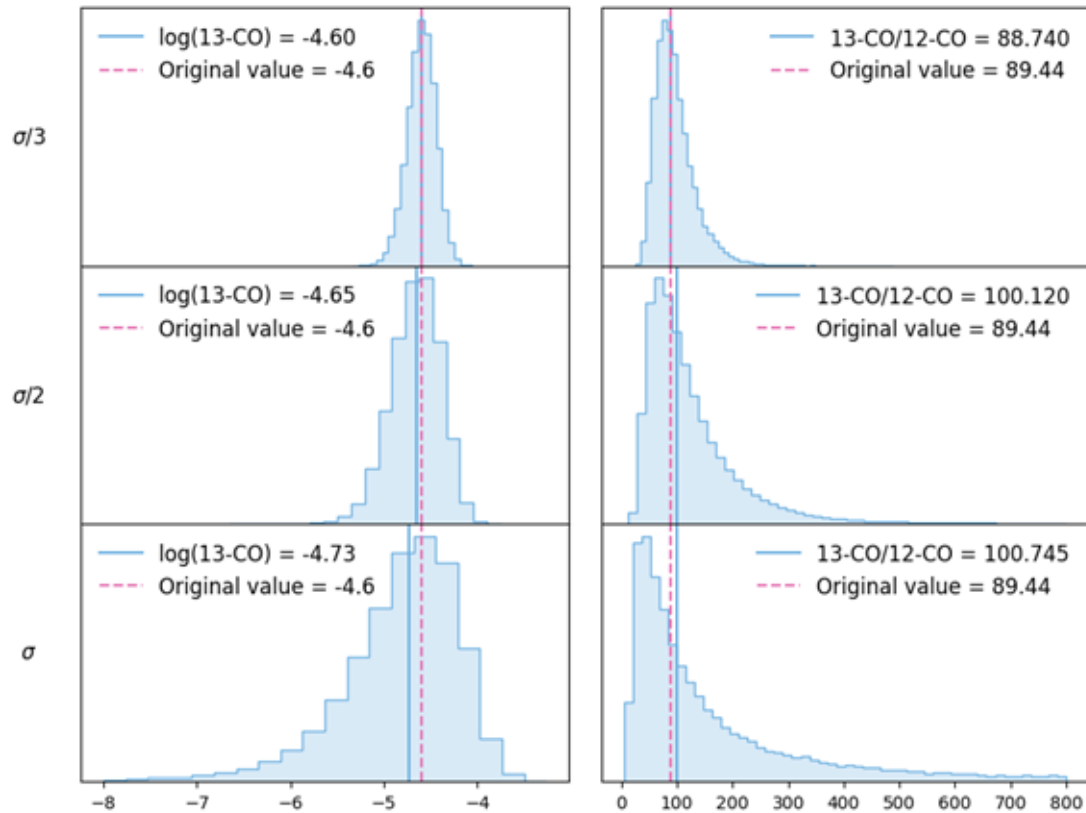
Why is the ratio biased to lower values?

Because $^{12}\text{CO}/^{13}\text{CO}$ is *inversely* proportional to ^{13}CO line strength

When we add Gaussian noise, the isotope posterior will be the inverted posterior of a Gaussian, which has a positive skew



Try fitting for log abundance directly $\rightarrow$ $\log(^{13}\text{CO})$ VMR

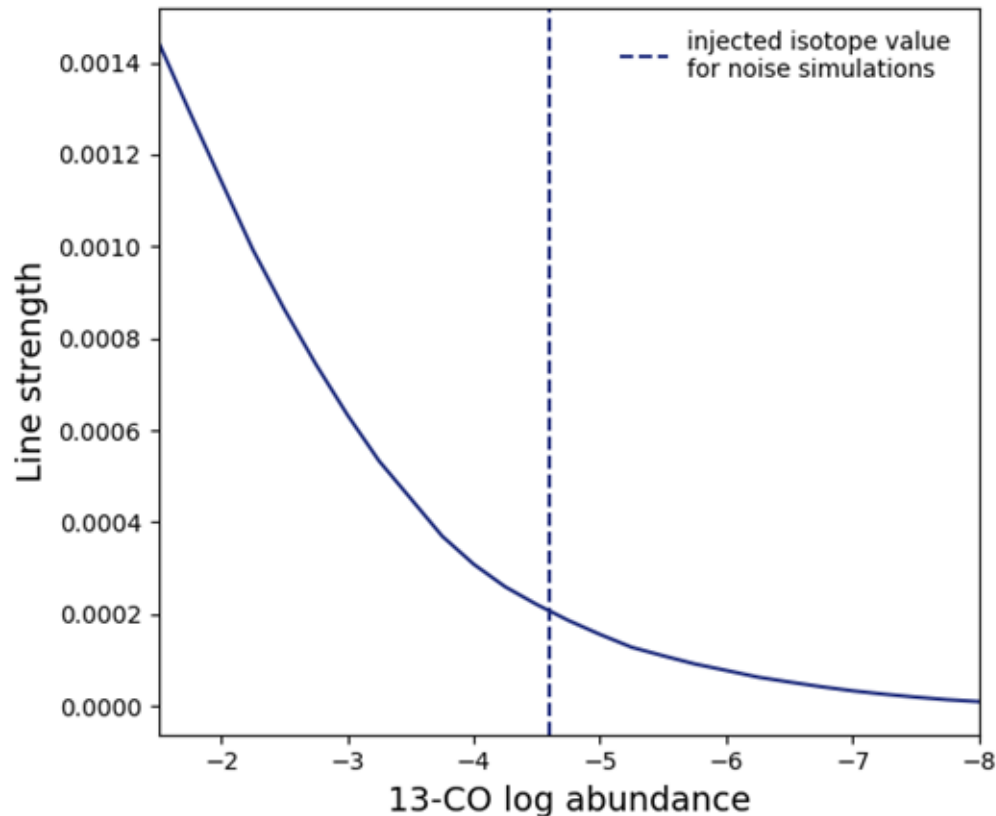


Why is the ratio biased to lower values?

Because of how ^{13}CO line strength scales and at low SNR

Samples more lower abundances than higher values (since the line strength doesn't change as much)

Bunching up near higher values $\rightarrow$ lower ratios



Running retrievals with log abundances

